

MMMB215 Project Submission 3

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Subject number and name:	MMMB215 – Mechanical Design 1
Subject Coordinator:	U [REDACTED] r
Title of Assignment:	Project Submission 3
Date and time due:	5/07/2026 11:59 PM
Lab Number:	11
Tutor Name:	A [REDACTED] d

Student declaration and acknowledgement

By submitting this assignment online, the submitting student declares on behalf of the team that:

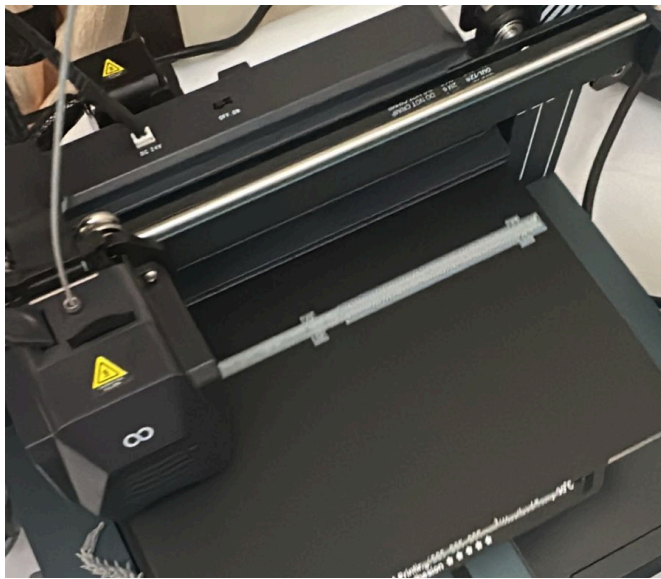
1. All team members have read the subject outline for this subject, and this assessment item meets the requirements of the subject detailed therein.
2. This assessment is entirely our own work, except where we have included fully documented references to the work of others. The material contained in this assessment item has not previously been submitted for assessment.
3. Acknowledgement of source information is in accordance with the guidelines or referencing style specified in the subject outline.
4. All team members are aware of the late submission policy and penalty.
5. The submitting student undertakes to communicate all feedback with the other team members.

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1. Construction of Design

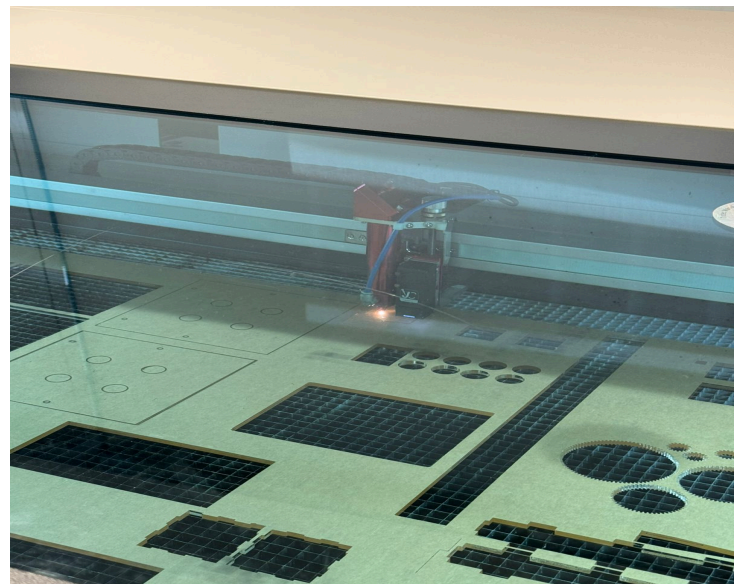
Shafts

To begin with, the basic elements of the gearbox were manufactured, starting with the shafts. Through trial and error, which will be further delved into, 3d printing was chosen to produce the shafts using 25% infill, using the ELEGOO Neptune Pro 4. These were printed horizontally as it structurally strengthens the shafts by being able to withstand the bending moments and shear stress experienced, since the extrusions run along the length.



Casing

Then, laser cutting was used to manufacture the gearbox casing. An acrylic sheet of 6mm was used. The Fusion360 drawings were formatted into .dxf files and then uploaded onto the software. Tolerance was tested first using a small sheet, in which bearing and press-fit holes of different diameters and widths were laser cut, and each was tested in order to figure out which diameter and width would be suitable for the final design. After that was initialised, the casing pieces were finally laser cut and press fit.



Gears

Finally, gears were 3d printed using 25% infill, using the ELEGOO Neptune Pro 4 and Creality K2 Plus with rafts for bed adhesion, which proved to be a difficulty to remove, hence when reprinting the gears, they were not used.

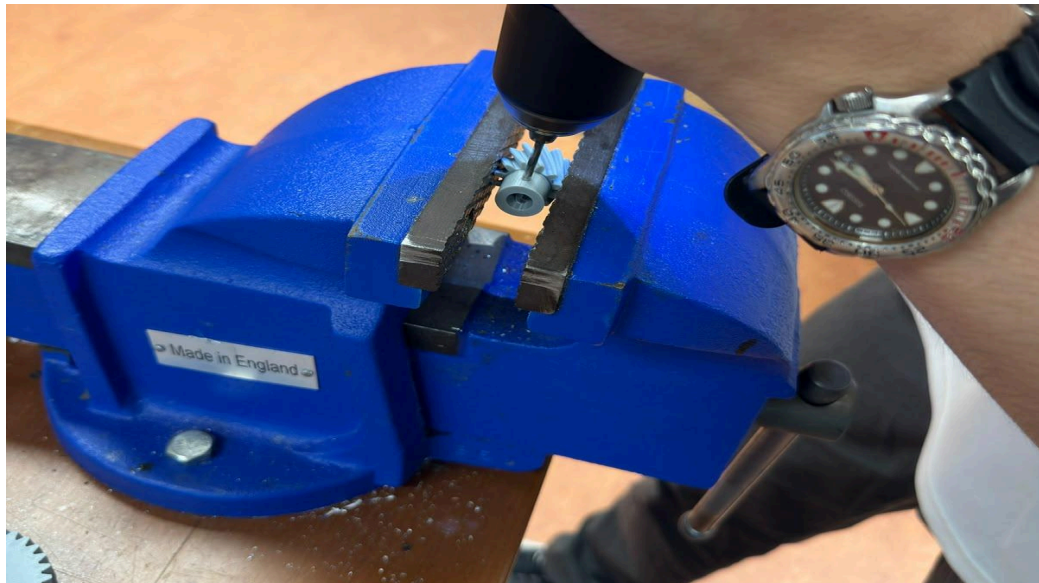


Assembly

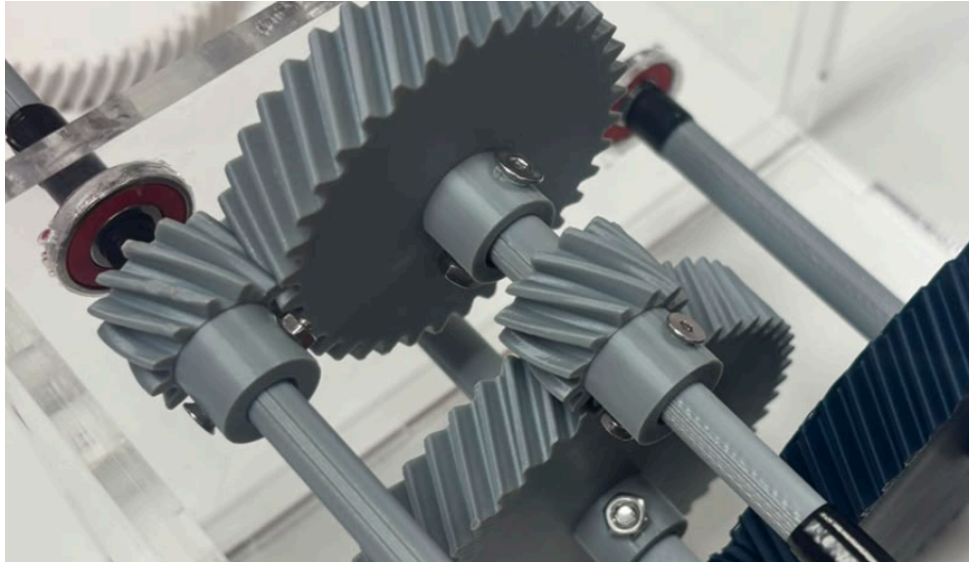
After manufacturing the casing, shafts and gears, they can all be assembled together into the final gearbox. The assembly process started by collecting the m3 and m4 bolts then threading them into their respective holes on the shaft. For instance, the holes that are 3mm wide on the shaft were threaded using the m3 bolt, and the 4mm hole was threaded using the m4 bolt. A tap was not needed for our case since the bolt itself acts as a tap that carves out its own thread by rotating it into the hole.



Though not strictly necessary, applying threading on the whole of the shaft where the bolt will rest will ensure that the gears will be securely locked to the shaft. Once the thread was applied, a hole now needed to be drilled onto the gear hub where the bolt and nut will keep the gear locked. Although a hole was already present and added onto the design before being 3D printed, the resulting hole was still too small for the bolt to go through, hence the need to drill a larger hole.



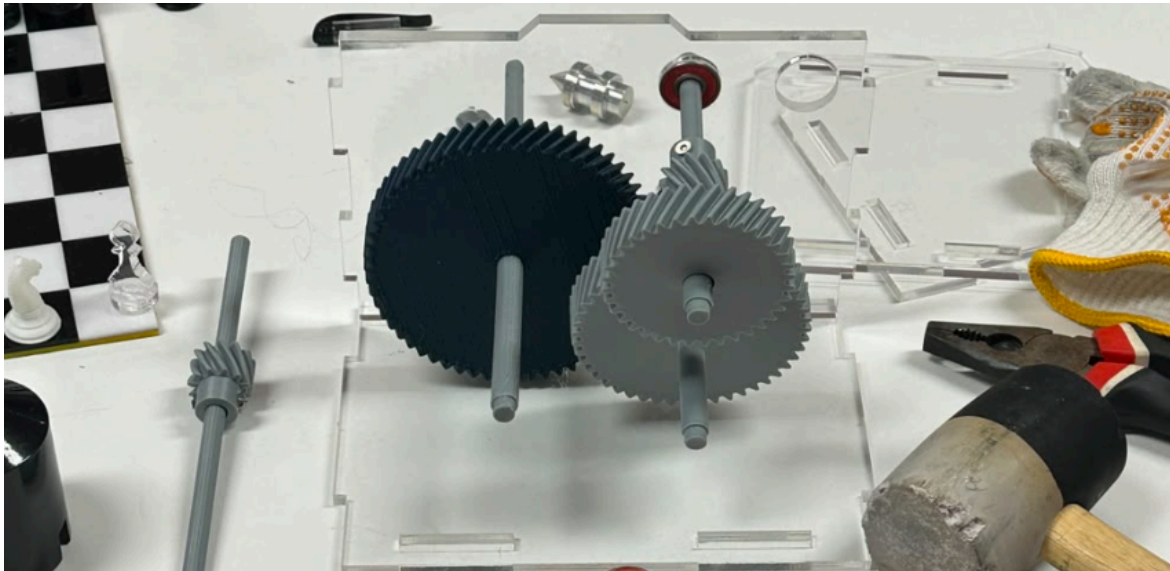
A 3mm drill bit was used for the m3 bolt and a 4mm drill bit for the m4 bolt. After the hole was sufficiently large enough, the bolt was placed through the holes of the gear hub and shaft, then secured together using the m3 and m4 nuts.



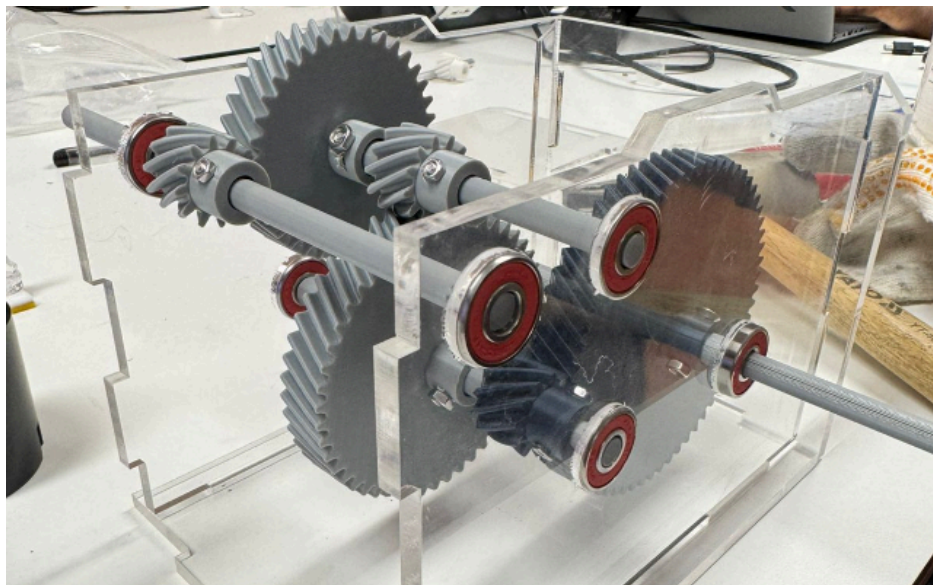
This was done to each shaft following the correct gear placement of CAD design. After all gears and shafts are connected, the next step will be to place the bearings into their designated slots on the casing.



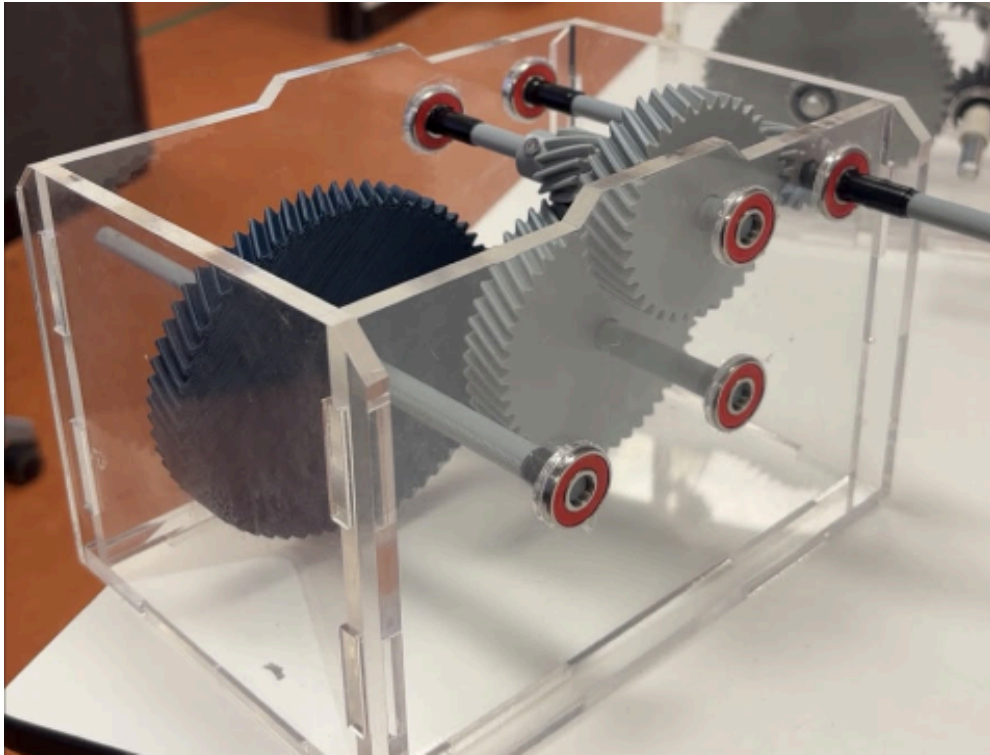
To keep the bearings from falling off, acrylic glue was used to secure the bearings in place. This ensured a tight and secured connection even whilst the shaft spins rapidly. Once all of the bearings are glued to their slots, the shafts can start to be placed onto the bearings. The base of the gearbox was first set in place, then the first wall casing was placed onto the base with a tight-press fit connection.



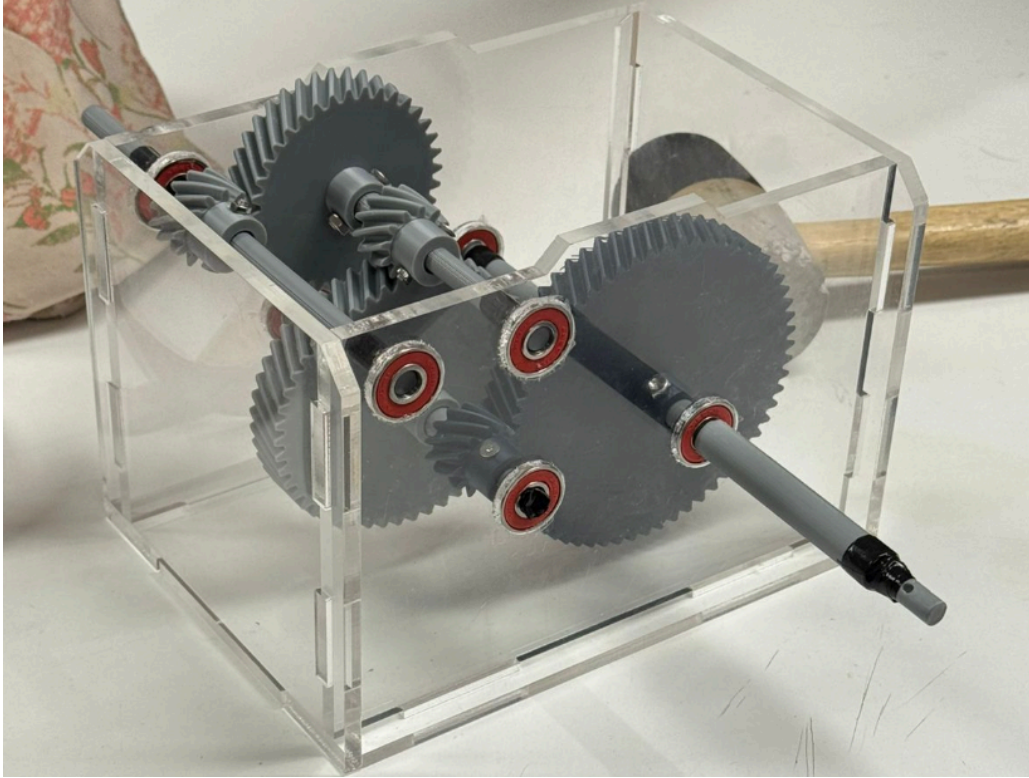
The next wall was then connected to the other end of the shaft, also ensuring a tight press-fit connection to the base.



After the main walls of the gearbox are secured into place, the sidewalls can now be placed. These are also press-fit and are connected to the base and the main walls, fastening everything in place and giving the entire gearbox its structural rigidity.



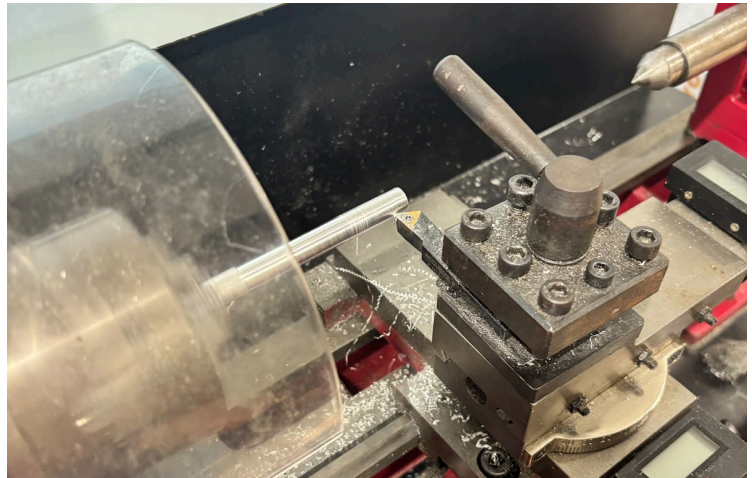
Lastly, to keep the shafts completely secure to the bearings and prevent vibrations as it spins, tape was added to the ends of the shafts, next to where they make contact with the bearings. This stops the shaft from sliding side to side and keeps it fastened. To finish off the design, it was noticed that the output shaft tended to bend under the 1 kilogram load that is placed on its end. To counteract this, an extra part was 3D printed that will go into the output shaft that adds an extra layer of thickness to the shaft. Tape was then wrapped around the inside output shaft layer and the extra part was taped onto the shaft to keep it in place. With this, the bending from the output shaft was significantly reduced and allowed the gearbox to easily support the 1 kilogram load. The final assembled gearbox is as shown:



The gearbox was tested for basic functionality and passed all metrics. The gears successfully intermeshed and transferred torque smoothly with no issues, the shafts can handle the load placed on them, and the casing proved to be secure and structurally rigid. Overall, the prototype works as intended for its basic functionality.

1. Challenges Faced

1. Initially, 10 mm aluminium shafts were sawn to the required length, then lathed to form the step shafts. However these were deemed unusable as the lathe machine was faulty and kept bending, resulting in uneven results. In addition, the digital measuring tool was not functional, hence the shafts diameter had to be measured after every rotation.
2. The output shaft was bending and not sturdy enough, hence requiring the printing of an extra part in order to reinforce and stabilize it.



3. The initial gear and shaft holes were too small, requiring reprinting and drilling in order for the bolts to fit. As well as the extra 3D printed piece for the output shaft having a tight fit, requiring sanding down. This was a tricky process as it was very prone to cracking, and testing to fit it onto the shaft would often push the bearing out of its place.
4. Material availability constraints necessitated a design change, shifting the wall thickness from 5mm to 6mm thick acrylic. As well as overall limited machine availability despite bookings due to the heavy demand, particularly with the 3D printers.

2. Finalised Budget (BOM)

Bill of Materials

Part	Material	Version	Notes / Change	Qty	PLA (g)	Aluminium Length (mm) (Ø10 mm rod)	Acrylic Area (mm ²) (6 mm)	Cost (Dhs)
Shaft - Input	Aluminium	v1		1	N/A	175	N/A	2.28
	PLA	v2	Switched to 3D printing. Changed dimensions.	1	7.65	N/A	N/A	0.92
Shaft - Stage 2	Aluminium	v1		1	N/A	125	N/A	1.63
	PLA	v2	Switched to 3D printing. Changed dimensions.	1	6.11	N/A	N/A	0.73
Shaft - Stage 3	Aluminium	v1		1	N/A	125	N/A	1.63
	PLA	v2	Switched to 3D printing. Changed dimensions.	1	6.58	N/A	N/A	0.79
Shaft - Output	Aluminium	v1		1	N/A	225	N/A	2.93
	PLA	v2	Switched to 3D printing. Changed dimensions.	1	11.09	N/A	N/A	1.33
Output Shaft Support	PLA	v1		1	5.47	N/A	N/A	0.66
Gear - Stage 1	PLA	v1		1	23.65	N/A	N/A	2.84
	PLA	v2	Changed dimensions.	1	23.57	N/A	N/A	2.83
Gear - Stage 2	PLA	v1		1	37.42	N/A	N/A	4.49
	PLA	v2	Changed dimensions.	1	37.31	N/A	N/A	4.48
Gear - Stage 3	PLA	v1		1	60.49	N/A	N/A	7.26
Pinion	PLA	v1		2	5.6	N/A	N/A	1.34
	PLA	v2	Changed dimensions.	3	5.62	N/A	N/A	2.02
Case - Bottom	Acrylic	v1		1	N/A	N/A	26671	3.89
Case - Front/Back	Acrylic	v1		2	N/A	N/A	23911	6.98
Case - Left/Right	Acrylic	v1		2	N/A	N/A	18422	5.38
608 Bearing	N/A	N/A		8	N/A	N/A	N/A	16.00
Bolts - M3	N/A	N/A		3	N/A	N/A	N/A	0.90
Bolts - M4	N/A	N/A		3	N/A	N/A	N/A	1.20
Nuts - M3	N/A	N/A		3	N/A	N/A	N/A	0.45
Nuts - M4	N/A	N/A		3	N/A	N/A	N/A	0.60

Total Cost (Dhs)	73.54
Budget (Dhs)	250.00
Remaining (Dhs)	176.46

3. Design Performance Reflection

To test the performance of our design, a string was tied to the end of the output shaft with the weight attached to the end. As the shaft rotates, the string will get pulled along with the weight. The length of the string that has been pulled will stay on the shaft as the weight gets lifted. In order for it to pass, it must lift the 1 kilogram mass by 120 centimeters within a 10 to 20 second window. The electric motor, which has an input rpm of 11000 was attached to the input shaft.



After testing, the time was taken to be roughly 14.5 seconds, which was within the designated window. The initial calculated time was 15 seconds, meaning the gearbox successfully lifted the weight and was accurate to the theoretical calculation. However, the slight change would have resulted from the increased shaft diameter that had to be placed onto the output shaft to counteract the bending, thus speeding up the lifting time by a small amount. Despite this, the difference is small and the required time limit was achieved.

The gearbox was observed to remain structurally sound with no issues in its integrity. No shaft or bearing came loose, the gears did not break, and the casing remained intact. Minimal vibrations were observed despite the high rpm that the shafts were experiencing.

Overall, the gearbox performed as expected and proved successful in accomplishing the designated task.

4. Future Improvements

1. Casing:

The casing for the gearbox was mostly optimal, with secured press-fit connections to the base and walls. However, there was still an issue with the press fit for the bearings, where they were too loose and required the use of acrylic glue to secure them. This can be improved by testing more tolerances of the bearing slots. Additionally, L-brackets can be used in future designs to prevent the need for tolerance testing with the laser cutter.

2. Shafts:

As for the shafts, the strength could be improved by increasing the infill percentage in order to reduce bending and increase tensile strength. While the tolerance was accounted for, the diameter was reduced at the point of contact with the bearings and required taping to secure them. This can be corrected by slightly increasing the diameter of the shafts at the bearings to ensure a proper press fit.

3. Gears:

The helical gears can be replaced by herringbone gears in order to cancel out axial force. Their manufacturing complexity would not pose an issue since the gears would be 3D printed. Additionally, the rafts can be reduced since they stuck to the gears and sanding them would reduce the gear ratio. Machine drilling can also be used instead of a manual drill in order to avoid peeling the gear hub filament and create a more aligned and consistent hole. Finally, each of the different 3D printed gears had different infill patterns, consistent infill patterns across all gears will allow for better consistency in strength.

4. Resource Availability and Machine Functionality

The unavailability of 5mm acrylic as well as the lathe malfunction caused a last minute design change for the casings and shafts. This can be avoided if the lab resources were confirmed beforehand.

5. Lab Bookings and Design Finalization

The heavy crowding of the lab caused a delay in the manufacturing and therefore an obstacle with testing the gearbox components beforehand. In the future, the design will be finalized earlier so that the lab bookings can be made without issues.

Contributions Sheet

There have been a number of instances where particular group members have not engaged in the project to an adequate level (sometimes making no contribution).

To take account of this, each group is required to list the percentage contribution of each group member (totaling 100%).

Each group member must sign this sheet in acknowledgment that they accept the contributions listed against them.

In the case where there are differences in the level of contribution by group members, the final individual group mark will be adjusted up or down.

Group number/name:

Group Member Name	Percentage Contribution
John [REDACTED]	25%
Jude [REDACTED]	25%
Noor [REDACTED]	25%
Shaik Ikramuddin Jeelani	25%